

BLOOD CANCER

Detailed Notes – Community Health Nursing | B.Sc. Nursing

1. Definition

Blood cancer is a group of cancers affecting the blood, bone marrow, lymphatic system, or blood-forming cells. The three major types are **leukemia, lymphoma and multiple myeloma**.

2. Major Types of Blood Cancer

Leukemia: Cancer of blood-forming cells in which abnormal cells, commonly white blood cells, multiply excessively and interfere with normal blood-cell production.

Lymphoma: Cancer of lymphocytes and the lymphatic system, commonly involving lymph nodes. It includes Hodgkin and non-Hodgkin lymphoma.

Multiple myeloma: Cancer of plasma cells in the bone marrow. It may cause bone damage, anemia, infection and kidney problems.

3. Types of Leukemia

- ALL – Acute Lymphoblastic Leukemia
- AML – Acute Myeloid Leukemia
- CLL – Chronic Lymphocytic Leukemia
- CML – Chronic Myeloid Leukemia

4. Risk Factors

- Increasing age
- Genetic/chromosomal abnormalities
- Family history in some blood cancers
- Previous chemotherapy or radiotherapy
- Significant radiation exposure
- Certain occupational chemical exposures, such as benzene
- Certain infections associated with specific blood cancers
- Some immune-system disorders

5. Pathophysiology – Simple Flow

Genetic changes/risk factors → abnormal blood-forming cell → uncontrolled multiplication → abnormal cells occupy bone marrow/lymphatic tissue → normal blood-cell production decreases → anemia + infection risk + bleeding → tissue/organ involvement.

6. Signs and Symptoms

- Persistent fever or recurrent infections
- Fatigue and weakness
- Pallor due to anemia
- Easy bruising or bleeding
- Nose or gum bleeding
- Unexplained weight loss
- Night sweats
- Painless lymph-node enlargement
- Bone or joint pain
- Loss of appetite

- Recurrent infections

7. Disease-wise Symptoms

Leukemia: fever, infections, anemia, bruising/bleeding, bone pain and lymph-node enlargement.

Lymphoma: painless lymph-node swelling, fever, night sweats, weight loss, fatigue and itching.

Multiple myeloma: bone pain, fractures, anemia, recurrent infections, high calcium and kidney problems.

8. Diagnosis

1. **CBC:** assesses hemoglobin, WBCs and platelets.
2. **Peripheral blood smear:** examines blood cells microscopically.
3. **Bone marrow aspiration/biopsy:** important for diagnosing many leukemias and myeloma.
4. **Lymph-node biopsy:** used when lymphoma is suspected.
5. **Imaging:** CT, MRI, PET-CT or X-ray as clinically indicated.
6. **Special tests:** flow cytometry, cytogenetic and molecular tests.

9. Treatment

Treatment depends on the exact type, stage/risk category and patient's condition.

- Chemotherapy
- Targeted therapy
- Immunotherapy
- Radiotherapy
- Hematopoietic stem-cell transplantation
- Blood or platelet transfusion and other supportive care

10. Community Health Nursing Management

A. **Primary Prevention:** Promote healthy lifestyle, balanced nutrition and physical activity; discourage tobacco; educate about occupational safety; reduce unnecessary radiation exposure; and promote awareness of cancer warning signs.

B. **Secondary Prevention:** Recognize persistent warning symptoms, encourage early medical consultation, facilitate appropriate referral, support recommended investigations, and follow up referred patients.

C. **Tertiary Prevention:** Promote treatment adherence, regular follow-up, infection and bleeding prevention, adequate nutrition, pain management, rehabilitation, psychological support and family education.

11. Nursing Care at Community Level

Infection prevention: hand hygiene, personal hygiene, safe food/water, avoiding exposure to infectious persons when immunity is low, temperature monitoring and prompt reporting of fever.

Bleeding prevention: avoid unnecessary trauma, observe for bruising/bleeding, use oral-care measures advised by the treating team, and promptly report significant bleeding.

Nutrition: balanced nutritious diet, adequate fluids unless restricted, small frequent meals when appetite is poor, and food-safety precautions.

Psychological care: listen to concerns, provide understandable information, encourage family support and refer for professional support when needed.

12. Health Education

- Take medicines and treatment exactly as prescribed.
- Do not stop cancer treatment without medical advice.
- Attend regular follow-up visits and investigations.
- Maintain hand hygiene and food safety.
- Avoid tobacco and unnecessary alcohol.
- Take adequate rest and maintain nutrition.
- Report fever, unusual bleeding, severe weakness, breathlessness or sudden deterioration promptly.

13. Role of Community Health Nurse

- Health education
- Risk-factor reduction
- Early recognition of warning signs
- Timely referral
- Treatment-adherence support
- Home visits and follow-up
- Monitoring for complications

- Infection and bleeding-prevention education
- Nutritional and psychosocial support
- Rehabilitation and family education

14. Prevention & Control Flow

Health education → Risk reduction → Early recognition → Timely referral → Diagnosis → Treatment adherence → Prevention of complications → Rehabilitation → Follow-up.

15. Important Exam Points

- Three major blood cancers: **Leukemia, lymphoma, multiple myeloma.**
- Common complications: **anemia, infection and bleeding.**
- Important investigations: **CBC, peripheral smear, bone marrow examination and disease-specific tests.**
- Community nursing focus: **prevention, early recognition, referral, follow-up, treatment adherence, complication prevention and health education.**

Conclusion

Blood cancer requires early recognition, accurate diagnosis and specialized treatment. Community health nurses play an important role in prevention, health education, early referral, home-based support, treatment adherence, complication prevention, rehabilitation and family counselling.